GROUP WORK PROJECT # ___
GROUP NUMBER: _____

MScFE 652: PORTFOLIO MANAGEMENT

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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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1. Introduction and data

A portfolio manager wants to build a portfolio for a client from ten large US stocks: Tesla (TSLA), Walmart (WMT), Bank of America (BAC), Goldman Sachs (GS), Eli Lilly (LLY), Merck (MRK), Alphabet class C (GOOG), Meta (META), Apple (AAPL) and ExxonMobil (XOM). In this report we work through four questions the manager asked us. First, what the optimal mean-variance mix is, with and without a cap on any single stock. Second, which style factors drive the capped portfolio. Third, whether random simulation can stand in for proper optimisation. Fourth, how the allocation changes once we add our own market views with the Black–Litterman approach.

We used **daily** closing prices, adjusted for splits, from 1 September 2023 to 30 September 2025 (521 daily returns) to estimate mean returns and the covariance matrix. The following quarter, 1 October to 31 December 2025 (64 trading days), was kept aside for out-of-sample testing. We chose daily over weekly data because two years of weekly data gives only about 105 observations for a 10×10 covariance matrix (55 parameters), while daily data gives five times as many and still covers the same market regime. Returns are simple daily returns and are annualised with 252 days. The risk-free rate is the daily one-month T-bill rate from the Kenneth French Data Library (French), the same source as the five factors used in Part 2 (Fama and French), so every excess return in the report is measured against the same benchmark. Its average over the sample was 5.04% a year. S&P 500 constituent weights on 30 September 2025 were used to build the market portfolio in Part 4

Table 1. In-sample return statistics, 1 Sep 2023 – 30 Sep 2025

Stock	Mean return (ann.)	Volatility (ann.)	Sharpe ratio
TSLA	45.8%	62.9%	0.65
WMT	33.3%	21.1%	1.34
BAC	31.7%	25.6%	1.04
GS	46.9%	28.3%	1.48
LLY	21.7%	35.2%	0.47
MRK	-9.7%	24.2%	-0.61
GOOG	32.1%	29.6%	0.91
META	50.4%	36.2%	1.25
AAPL	18.5%	27.7%	0.49
XOM	3.1%	22.2%	-0.09

Table 1 shows how uneven the period was. META, GS and TSLA returned more than 45% a year, while Merck lost almost 10% and Exxon barely beat cash. Any optimiser fed these numbers will chase the winners, which is the main issue we keep coming back to below (Michaud)

2. Technical report

2.1 Part 1 – Standard mean-variance optimisation (Steps 1–4)

Method Following Markowitz, we look for the portfolio with the highest Sharpe ratio (Sharpe), i.e. the tangency portfolio, by solving the constrained problem below directly with a sequential quadratic programming solver. We used several starting points to make sure we were finding the global optimum and not a local one. No simulation was used in this part.

$$\max_{w} \frac{w^T\mu - r_f}{\sqrt{w^T\Sigma w}} \quad \text{s.t.} \quad \sum_{i=1}^{10} w_i = 1, \qquad 0 \le w_i \le u \tag{1}$$

Here mu is the vector of mean daily returns, Sigma the sample covariance matrix and r_f the average daily risk-free rate. In Step 1 the upper bound is u = 1 (long-only). For Step 2 the brief gives the client limit as 15% in one sentence and as 0.18 in the next, so we solved both cases. We treat **15%** as the binding mandate because it is what the client actually asked for, and a 15% portfolio automatically satisfies an 18% limit as well. The 15% portfolio is the one carried into Parts 2 and 4.

Table 2. Optimal (maximum Sharpe ratio) weights and in-sample statistics

Stock	Step 1: long-only	Step 2: max 15%	Step 2: max 18%
TSLA	0.0%	1.6%	0.0%
WMT	43.7%	15.0%	18.0%
BAC	0.0%	15.0%	18.0%
GS	34.7%	15.0%	18.0%
LLY	0.0%	14.4%	10.0%
MRK	0.0%	0.0%	0.0%
GOOG	4.9%	15.0%	18.0%

Stock	Step 1: long-only	Step 2: max 15%	Step 2: max 18%
META	16.7%	15.0%	18.0%
AAPL	0.0%	4.1%	0.0%
XOM	0.0%	4.9%	0.0%
Exp. return	40.8%	34.0%	37.2%
Volatility	19.2%	18.4%	19.1%
Sharpe ratio	1.86	1.57	1.69
Stocks held	4	9	6

Step 1 and Step 2 results

When we don't put any limit on how much each stock can get, the optimiser only picks four stocks: Walmart (43.7%), Goldman Sachs (34.7%), Meta (16.7%) and a small 4.9% in Alphabet. Michaud warns about exactly this. He calls it "error-maximising," because the optimiser just piles money into whichever stocks had the best Sharpe ratio in the past (the best return for the risk taken), as long as those stocks don't move too much like each other (Michaud). Walmart gets the biggest share because its price didn't jump around much (21% volatility) and it didn't really move with the tech stocks or the banks. Based on the past data we used, the Sharpe ratio comes out at 1.86.

When we add a rule that no stock can be more than 15%, the money has to be spread across nine stocks. Walmart, Bank of America, Goldman Sachs, Alphabet and Meta all hit the 15% limit. Eli Lilly gets 14.4%, and the last 10% or so is shared between Exxon, Apple and Tesla. Merck is the only stock that gets nothing. The Sharpe ratio falls to 1.57, so on paper the limit costs us about 0.29. When we set the limit at 18% instead, the portfolio holds six stocks and the Sharpe ratio lands in the middle, at 1.69.

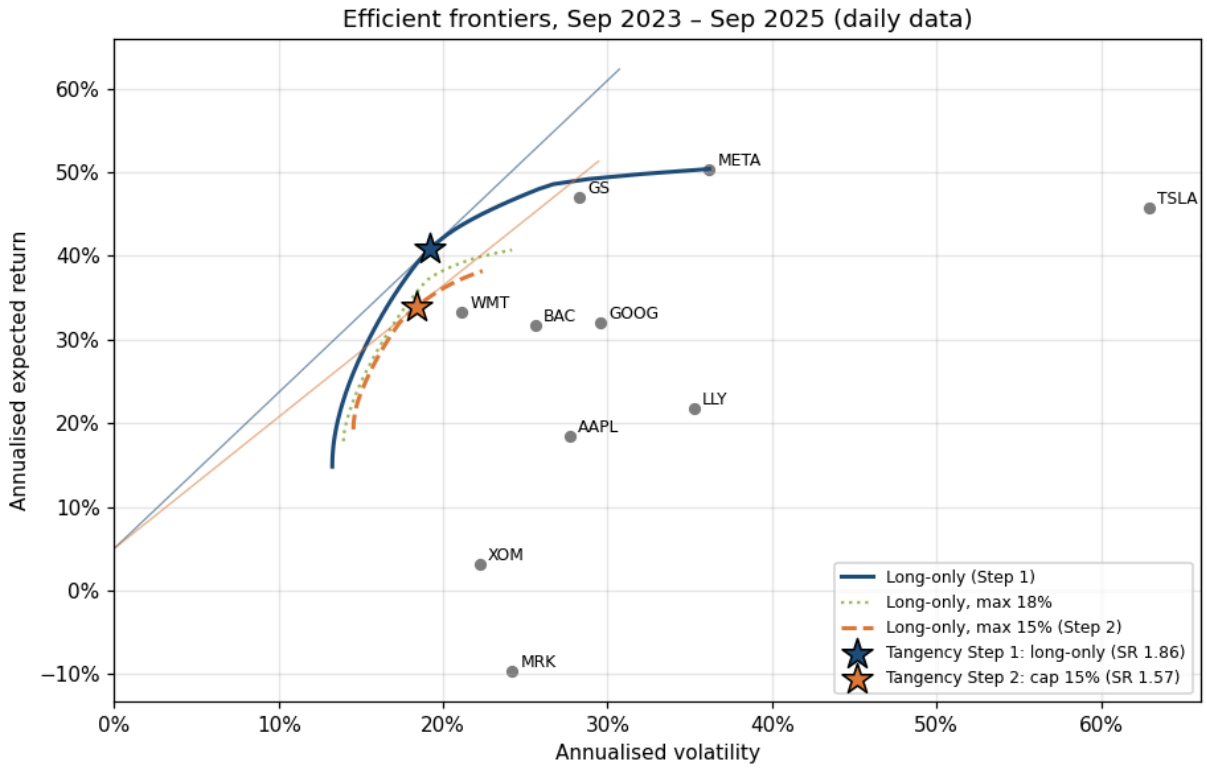


Figure 1. Efficient frontiers (long-only, 18% cap and 15% cap) with tangency portfolios and capital market lines

Step 3 – Comparing the efficient frontiers

Figure 1 puts all three frontiers on the same chart. The capped frontier is a lot shorter than the one with no limits, and it sits inside it. At the top, the long-only frontier (no short selling) reaches a 50.4% expected return, which just means putting 100% of the money in Meta. With the 15% limit, the highest we can reach is 38.2%, because no single high-return stock is allowed to take more than 15%. The bottom of the curve changes too. The lowest-risk portfolio goes from 13.3% to 14.6% volatility. This happens because the limit also stops us from putting a lot of money into the safer, low-risk stocks like Walmart, Exxon and Merck. The two curves only get close in the middle, where the best portfolios without limits already stay under 15% in each stock anyway.

This is what we expected. When you add a rule, you can only end up with fewer possible portfolios, so the capped frontier can never sit above the original one. In real terms, the cost shows up as a flatter capital market line and a lower best possible Sharpe ratio.

Step 4 – Out-of-sample performance (Oct–Dec 2025)

We bought each portfolio at the closing prices on 30 September 2025 and just held it until 31 December 2025, without changing the weights along the way. For each portfolio we show three things: the total return over the period, the Sharpe ratio (worked out from daily returns above the risk-free rate and then scaled up to a yearly number), and the maximum drawdown, which is the biggest drop from any earlier high point.

Table 3. Out-of-sample performance, 1 Oct – 31 Dec 2025 (buy-and-hold)

Portfolio	Cumulative return	Sharpe ratio (ann.)	Max drawdown	Volatility (ann.)
Step 1: long-only	6.87%	1.51	-5.02%	14.7%
Step 2: cap 15%	13.09%	3.42	-2.97%	12.9%
Step 2: cap 18%	11.97%	2.95	-2.91%	13.7%
Equal weight (1/10)	12.47%	3.15	-3.61%	13.4%
S&P 500	2.35 %	0.39	-5.11%	12.6%

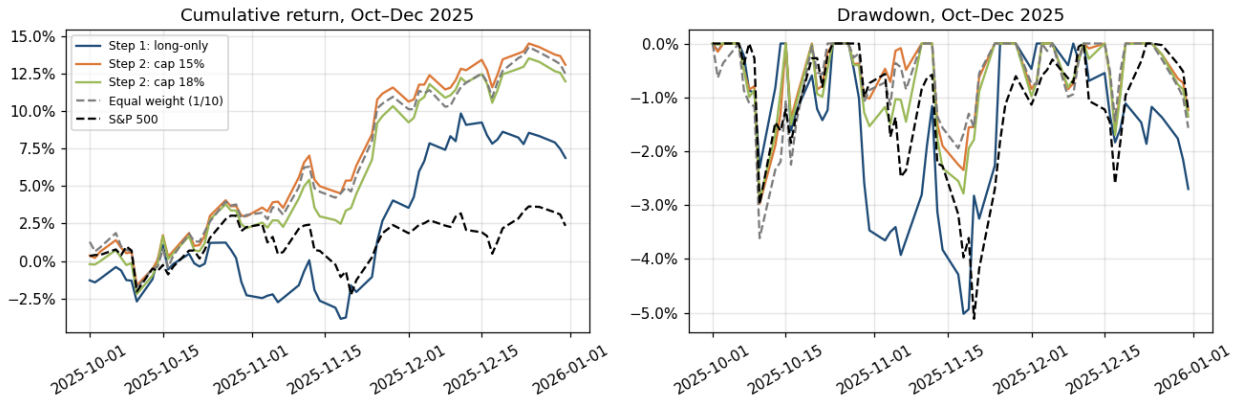


Figure 2. Out-of-sample cumulative return and drawdown

When we test on new data (the period after the one we used to build the portfolios), the ranking turns upside down. The capped portfolio made 13.1% over the three months, with a Sharpe ratio of 3.42 and a biggest drop of only –3.0%. The portfolio with no limits made 6.9%, with a Sharpe ratio of 1.51 and a biggest drop of –5.0%. Both beat the S&P 500, which went up 2.3%.

Most of the difference between our two portfolios comes from the stocks the concentrated one didn't hold. Eli Lilly went up 41% in the quarter, Alphabet 29% and Merck 25%, and the portfolio with no limits had very little or nothing in any of them. On top of that, Meta, its third-biggest holding, fell 10%. We were also surprised that the simple equal-weight portfolio (putting the same amount in every stock) made 12.5% and beat the "optimal" portfolio with no limits. This matches what Jagannathan and Ma found. They show that putting a limit on weights works a bit like toning down our guesses for returns and risk, which are often wrong, so the limits tend to help when we test on new data (Jagannathan and Ma). Three months is a short time, though, and a Sharpe ratio above 3 shouldn't be trusted too much. Still, the result points in the same direction the theory says it should.

2.2 Part 2 – Fama–French five-factor exposure of the capped portfolio (Steps 4–6)

Step 4 – The five factors and their correlations

We run the Fama and French five-factor model on the daily returns of the Step 2 portfolio (its returns above the risk-free rate), keeping the portfolio weights the same the whole time:

$$R_{p,t} - R_{f,t} = \alpha + \beta_1(R_{m,t} - R_{f,t}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + \varepsilon_t \quad (2)$$

- **Market (Mkt–RF)**: the return of the whole US stock market (bigger companies count more) minus the return on T-bills. Its beta shows how much the portfolio moves when the overall market moves. This is the kind of risk you can't get rid of just by owning more stocks.
- **SMB (Small Minus Big)**: the return of small companies minus the return of big companies. A positive beta means the portfolio leans towards small companies. A negative beta means it acts more like big companies.
- **HML (High Minus Low)**: the return of "value" stocks (cheap compared to their book value) minus the return of "growth" stocks (expensive compared to their book value). A positive beta means the portfolio leans towards value stocks.
- **RMW (Robust Minus Weak)**: the return of companies that make strong profits minus the return of companies with weak profits. A positive beta means the portfolio leans towards solid, profitable companies (Fama and French).

- **CMA (Conservative Minus Aggressive)**: the return of companies that grow their assets slowly minus the return of companies that grow their assets fast. A negative beta means the portfolio leans towards companies that are spending a lot to grow and expand (Fama and French).

Table 4. Correlation matrix of daily factor returns, Sep 2023 – Sep 2025

Factor	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.00	0.21	-0.26	-0.38	0.02
SMB	0.21	1.00	0.39	-0.40	0.15
HML	-0.26	0.39	1.00	0.19	0.10
RMW	-0.38	-0.40	0.19	1.00	-0.11
CMA	0.02	0.15	0.10	-0.11	1.00

None of the factors move very closely together. The strongest links are SMB with RMW (–0.40), SMB with HML (0.39) and the market with RMW (–0.38). The negative link between the market and RMW makes sense for this period. The market's rise was driven by a few very large tech companies, but on big up days, small and less profitable companies also tended to jump, and that pulled RMW down. CMA barely moves with any of the other factors, and the market factor only has a weak link with SMB and HML. We also checked the variance inflation factors (a test for whether the factors overlap too much), and they were all below 1.8. So overlap between the factors isn't a problem here, and we can look at each beta separately.

Step 5 – OLS and robust regression; training and testing split

Splitting the data. Because our data is a time series (it follows dates in order), we split it by date instead of randomly. If we split it randomly, the model could learn from days that come after some of the days we test it on. That's like peeking into the future, which is called look-ahead bias. Our training data goes from 1 September 2023 to 30 September 2025 (521 days), which is the same period we used to choose the portfolio weights. The test data is the next three months, 1 October to 31 December 2025 (64 days). We work out the coefficients using only the training data. Then we take the actual factor returns from the test period, use them to predict the portfolio's returns above the risk-free rate, and compare the predictions with what really happened. As an extra check, we did the whole thing again with an 80/20 split by date inside the training period (training on Sep 2023 to Apr 2025, testing on May to Sep 2025).

Models. The first model is ordinary least squares (OLS) with Newey–West standard errors using 5 lags (Newey and West). Daily returns don't have the same amount of ups and downs every day, and today's return can be a little linked to yesterday's. Because of this, normal OLS standard errors would make our results look more certain than they really are. The second model is a robust M-estimator using Huber's loss, with the standard setting of 1.345 (Huber). It treats small errors the same way OLS does, but it gives less weight to big ones. We wanted this because a few big jumps on earnings days, like Tesla or Meta moving 10% in one day, could otherwise drag the betas around too much. In the end, about 22% of the training days were given less than full weight.

Table 5. Five-factor regression of the capped portfolio's daily excess return (training set)

Factor	OLS coef.	OLS t (HAC)	OLS p	Huber coef.	Huber t	Huber p
Alpha (ann.)	0.073	1.38	0.169	0.086	1.73	0.083
Mkt-RF	1.046	40.43	0.000	1.041	48.39	0.000
SMB	-0.157	-3.86	0.000	-0.150	-4.56	0.000
HML	0.202	4.46	0.000	0.190	5.57	0.000
RMW	-0.095	-1.64	0.102	-0.069	-1.52	0.128
CMA	-0.335	-4.54	0.000	-0.276	-6.03	0.000

Table 6. In-sample and out-of-sample fit (RMSE of daily excess returns)

Model	Train R2	Test R2	Train RMSE	Test RMSE	Test correlation
OLS	0.819	0.475	0.492%	0.578%	0.787
Huber robust	0.819	0.513	0.493%	0.556%	0.792

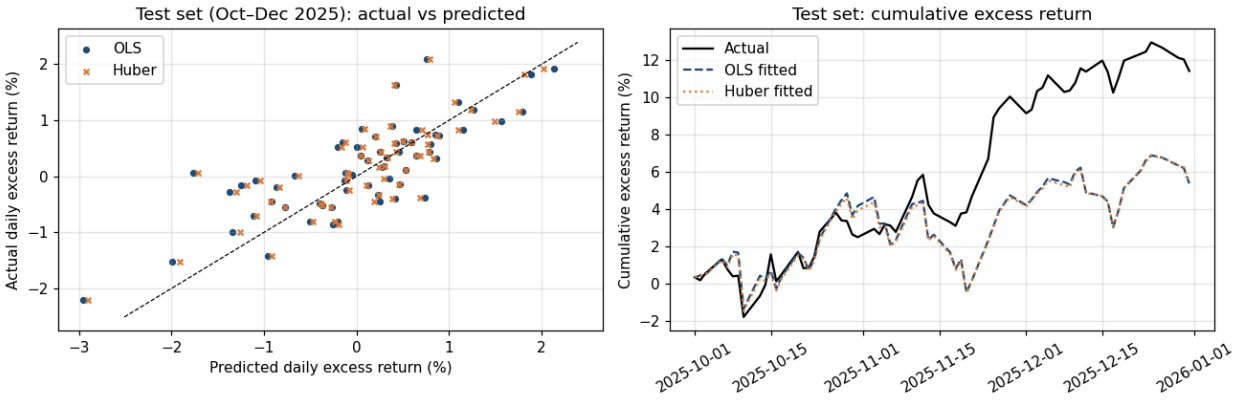


Figure 3. Test set (Oct–Dec 2025): actual vs predicted daily excess returns and cumulative excess return

Step 6 – Which factors explain the portfolio, and how

OLS and Huber give very similar results, so we're fairly sure the answers aren't being caused by just a few extreme days.

- **Market**: This is by far the most important factor. The beta is about 1.05, which means the portfolio takes on a bit more market risk than the index. This is mostly because of Tesla, Meta and the two banks. The market factor alone explains most of the 82% R^2.
- **Size**: The portfolio clearly leans towards big companies. That's no surprise, because all ten stocks are huge companies. Small companies did worse over this period (SMB was about –8% a year on average), so leaning towards big companies actually added around 1.3% a year.
- **Value**: There's a moderate lean towards value stocks, which comes mostly from Bank of America, Goldman Sachs, Exxon and Walmart. This partly balances out the growth lean from the tech stocks.
- **Investment**: The portfolio has a negative CMA beta, so it acts like a group of companies that spend a lot to grow. That makes sense, since Alphabet, Meta and Lilly are all spending heavily on AI data centres or new factories. The Huber number is a bit smaller than the OLS one, which tells us part of this comes from days with big price moves.
- **Profitability**: We expected a positive RMW beta, because most of these companies make a lot of profit. But the number turned out small and not statistically significant. Our guess is that part of the profitability effect is already being picked up by the market factor, since the two have a –0.38 correlation.
- **Alpha**: 7.3% a year with OLS (p=0.17) and 8.6% with Huber (p=0.08). Neither one is significant at the 5% level, so we can't say we have any real skill at picking stocks. Most of the extra return is explained by the factors.

Table 7. Return attribution of the capped portfolio (training set)

Factor	Beta (OLS)	Avg factor return (ann.)	Contribution (ann.)
Mkt-RF	1.046	17.0%	17.81%
SMB	-0.157	-8.1%	1.26%
HML	0.202	2.4%	0.49%
RMW	-0.095	-2.6%	0.25%
CMA	-0.335	-5.4%	1.82%
Alpha	—	—	7.28%
Total			**28.92%**

Table 7 splits the 28.9% average yearly excess return into the pieces that make it up. The market gives 17.8 points. Size adds 1.3, investment 1.8, value 0.5 and profitability 0.3, so together these add up to about 4 points. The last 7.3 points are alpha, the part the factors can't explain. Something we found interesting is that the negative SMB and CMA betas actually helped returns, because both of those factors lost money over this period. The downside is that if small companies bounce back, or if slow-growing, low-spending companies become popular again, the portfolio would fall behind the market.

When we test on new data, the model explains about half of the day-to-day changes in returns. The test R^2 is 0.47 for OLS and 0.51 for Huber, and the correlation between predicted and actual returns is about 0.79. Most of the drop from the 0.82 we got on the training data happens because the portfolio's biggest winners in Q4 were the healthcare stocks, Lilly and Merck. Their prices jumped because of company-specific news, which the factors can't capture. Huber does slightly better on the test data, which we think is a good reason to use the robust numbers when reporting risk. On the 80/20 split inside the training period, the betas hardly changed and the test R^2 was 0.68 for both models. So the portfolio's factor exposures look steady over time.

2.3 Part 3 – Can simulation replace optimisation? (Steps 7–8)

We picked five stocks from five different sectors so the problem wouldn't be too easy: Alphabet (communication services), Eli Lilly (health care), Walmart (consumer staples), Goldman Sachs (financials) and Exxon (energy). Like before, we look for the weights that give the highest Sharpe ratio, with the weights adding up to 100% and no short selling. In Step 8 we add a limit of 30% per stock.

How the simulation works. We create random portfolio weights using a Dirichlet distribution with all its settings equal to one. This spreads the random portfolios evenly across every possible way of splitting 100% between the five stocks (with no short selling), so every portfolio has the same chance of being picked. We didn't use the more common method of drawing five random numbers and dividing each one by their total. That method tends to give weights close to equal, and it rarely gives portfolios that are heavy in just one or two stocks. The problem is that the best portfolio is often exactly that kind of lopsided one. For each number of random portfolios N, we keep the one with the highest Sharpe ratio and check the biggest gap between its weights and the optimiser's weights. One single run could be lucky or unlucky, so we repeated each N 30 times using different random seeds (10 times for N = 500,000) and took the average.

The 30% limit in Step 8 makes the simulation harder, because a random portfolio only counts if all five weights are 30% or less. For a random portfolio spread evenly across all possible five-stock splits, the chance of that happening is:

$$P\left(\max_i w_i \le c\right) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (1-kc)_+^{n-1} \tag{3}$$

$$= 1 - 5(0.7)^4 + 10(0.4)^4 - 10(0.1)^4 \approx 0.0545 \qquad (n = 5,\ c = 0.3)$$

So only about one random portfolio out of every 18 can actually be used. In our own runs, 5.49% passed the limit, which matches the math. We dealt with this using rejection sampling, which just means we create random portfolios and throw away any that go over the 30% limit. The ones we keep are still spread evenly across all the portfolios that meet the limit. This way we can compare Steps 7 and 8 fairly, because N only counts the portfolios that follow the rule.

Table 8. Optimal weights: optimisation vs best of 1,000,000 simulated portfolios

Stock	Step 7 optimiser	Step 7 simulation	Step 8 optimiser	Step 8 simulation
GOOG	12.5%	10.5%	29.1%	27.7%
LLY	0.6%	1.3%	10.9%	12.0%
WMT	47.5%	48.8%	30.0%	29.9%
GS	39.3%	39.3%	30.0%	29.7%
XOM	0.0%	0.1%	0.0%	0.7%
Sharpe ratio	**1.798**	**1.797**	**1.684**	**1.680**

Table 9. Convergence of simulation to the optimiser (average over repeated runs)

N (feasible)	Step 7: max weight error	Step 7: Sharpe gap	Step 8: max weight error	Step 8: Sharpe gap	Step 8: raw draws
100	13.18%	0.0760	8.87%	0.0739	2,036
500	9.42%	0.0373	5.98%	0.0381	10,010
1,000	7.08%	0.0240	5.25%	0.0281	20,010
5,000	4.83%	0.0123	3.89%	0.0172	100,010
10,000	4.31%	0.0092	3.30%	0.0142	200,000
50,000	2.93%	0.0044	2.04%	0.0074	928,540
100,000	2.64%	0.0033	1.55%	0.0056	1,839,195
500,000	2.47%	0.0020	1.15%	0.0032	9,190,413
1,000,000	1.81%	0.0015	1.44%	0.0028	18,356,586

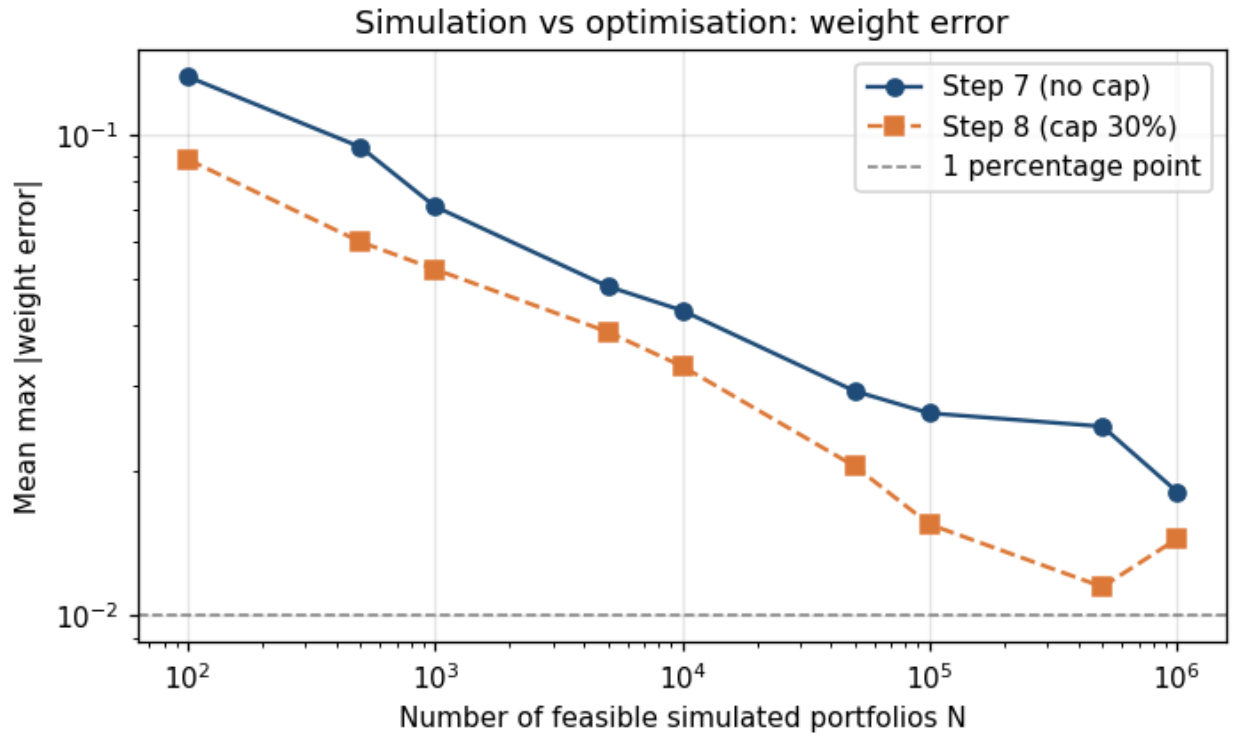


Figure 4. Average largest weight error vs number of simulated portfolios (log–log scale)

Step 7 result. The optimiser's answer is 47.5% Walmart, 39.3% Goldman Sachs, 12.5% Alphabet, 0.6% Eli Lilly and nothing in Exxon. The best Sharpe ratio from the simulation gets close to the real best one quickly, within 0.01 after about 10,000 random portfolios. The weights are a different story, though. They get close very slowly. The error goes down roughly in line with $1/\sqrt{N}$: about 13 percentage points at 100 portfolios, 4–5 points at 5,000–10,000, and still 1.8 points on average at one million. This happens because the Sharpe ratio barely changes near

the top, so portfolios with quite different weights can end up with almost the same score. In our runs, it took about one million simulations before all five weights were reliably within 2 points of the optimiser, and even then only some runs got within 1 point.

Step 8 result. With the 30% limit, Walmart and Goldman Sachs both sit at exactly 30%. The weight they lose goes to Alphabet (29.1%) and Eli Lilly (10.9%), and Exxon still gets nothing. If we only count the portfolios that follow the limit, the capped problem gets close to the answer about as fast as the uncapped one, maybe a little faster. That's because there are fewer possible portfolios, and two of the weights are stuck right at the limit. The problem is that each usable portfolio takes about 18 random tries to find. To get within about 1–1.5 points, we needed 500,000 to 1,000,000 usable portfolios, which means 9 to 18 million random tries in total. The optimiser finds the exact answer in less than a second.

What we learned is that simulation is useful for seeing all the possible portfolios and for double-checking the optimiser. But it shouldn't replace the optimiser, especially when there are limits on the weights. The number of wasted tries goes up fast as we add more stocks or make the limit tighter.

2.4 Part 4 – Black–Litterman (Step 9)

(a) Market portfolio and its Sharpe ratio

The Black-Litterman model (Black and Litterman) needs a market portfolio to start from, and ten stocks by themselves aren't a whole market. To get around this, we built a made-up combined portfolio. We use the S&P 500 as the market. Our ten stocks are included at their real weights in the index on 30 September 2025. For Alphabet, we only use the class C shares, because GOOG is the one we hold. We then add an eleventh asset called "REST," which stands for the other roughly 490 companies in the index and gets whatever weight is left over, 1 - sum w_i = 82.2% . We work out the daily return of REST from the index's return:

$$r_{REST,\,t} = \frac{r_{SPX,\,t} - \sum_{i=1}^{10} w_i r_{i,\,t}}{w_{REST}} \qquad (4)$$

We then work out Σ, the covariance matrix, for all eleven assets. It shows how much each asset moves and how they move together. As a check, $w^{\top}\Sigma w$ gives back exactly the S&P 500's variance (0.02496 per year). One weakness is that we keep the September 2025 weights the same for the whole two years, even though in real life index weights change over time.

Over this period, the S&P 500 made 15.3% a year above the risk-free rate, with 15.8% volatility, so the market Sharpe ratio is 0.97. From this we get the risk-aversion number, which shows how much extra return investors want for taking on more risk:

$$\delta = (E[R_m] - R_f)/\sigma_m^2 = SR/\sigma_m = 6.13$$

We then work backwards from the market weights to find the "equilibrium" returns, meaning the returns the market is quietly assuming for each asset. This is called reverse optimisation (Black and Litterman; He and Litterman):

$$\Pi = \delta \Sigma w_{mkt} \quad (5)$$

Table 10. Market weights (30 Sep 2025) and equilibrium vs historical excess returns (annualised)

Asset	Market weight	Implied excess return Π	Historical excess return
TSLA	2.19%	35.5%	40.8%
WMT	0.80%	8.2%	28.3%
BAC	0.63%	15.1%	26.7%
GS	0.43%	20.1%	41.9%
LLY	1.02%	11.9%	16.7%
MRK	0.35%	4.7%	-14.7%
GOOG	2.01%	17.2%	27.0%
META	2.84%	22.5%	45.4%
AAPL	6.65%	18.3%	13.5%
XOM	0.86%	6.4%	-1.9%
REST	82.24%	14.4%	13.4%

The implied returns are much more sensible as forecasts than the historical averages. Merck and Exxon get small positive premia instead of negative ones, and Meta, Goldman and Walmart are pulled down to what their risk justifies. Tesla gets the highest prior (35.5%) only because its beta is very high. Our delta of 6.1 is above the usual 2.5–3.5 range because the 2023–2025 market Sharpe ratio was unusually high. We kept it so that the prior is internally consistent with the data, but note that it makes the implied premia relatively generous.

(b) Choice of tau

tau measures how uncertain we are about the equilibrium mean (Walters), through Pi sim N(mu, tau \Sigma). If means are estimated from T observations, the covariance of the sample mean is \Sigma, so we set **tau = 1/T = 1/521 = 0.00192$**. This choice is consistent with the data frequency. With daily data, tau Sigma is the variance of an estimated *daily* mean. A value like 0.025–0.05, often quoted for monthly or annual data (Walters), would be far too large here and would make the prior almost meaningless. All inputs (Pi, Q,Omega) are kept in daily units and only annualised for presentation.

(c) Views: P, Q and Omega

We took the role of the portfolio manager on 30 September 2025 and wrote four views:

- **View 1 (relative, high confidence): LLY will outperform MRK by 8% a year**. Lilly's GLP-1 drugs (Mounjaro, Zepbound) are still growing very fast, while Merck faces the loss of exclusivity on Keytruda from 2028.
- **View 2 (absolute, medium confidence): XOM will earn 0% over cash**. OPEC+ has been unwinding production cuts through 2025 and Brent is trading in the mid-$60s, which squeezes upstream earnings.
- **View 3 (relative, medium confidence): GOOG will outperform META by 5% a year**. The September 2025 search antitrust remedy turned out much milder than feared (no forced sale of Chrome), and Google Cloud and Gemini are monetising AI, while Meta keeps raising its capex guidance without a matching revenue line.
- **View 4 (relative, low confidence): an equal-weighted pair of BAC and GS will outperform the rest of the market by 3% a year**. This rests on a steeper yield curve, lighter capital rules (the proposed eSLR changes) and a recovery in M&A and IPO activity.

Table 11. Pick matrix P (columns with non-zero entries; all other assets are 0)

View	BAC	GS	LLY	MRK	GOOG	META	XOM	REST
V1: LLY - MRK	0	0	1	-1	0	0	0	0
V2: XOM	0	0	0	0	0	0	1	0
V3: GOOG - META	0	0	0	0	1	-1	0	0
V4: Banks - REST	0.5	0.5	0	0	0	0	0	-1

Each row of P is one view, and every asset not shown in the table has a zero in that row. The view vector is:

$$Q_{annual} = (8\%,\ 0\%,\ 5\%,\ 3\%)^T, \qquad Q_{daily} = Q_{annual}/252 \tag{6}$$

Uncertainty Omega. We start from He and Litterman's rule, which sets each view's variance equal to the prior variance of the same portfolio (He and Litterman), and then scale it by a confidence multiplier c_k: 0.5 for high confidence (half the variance), 1 for medium and 2 for low. Omega is diagonal, so the views are treated as independent.

$$\Omega = \mathrm{diag}(c_k\,[P(\tau\Sigma)P^T]_{kk}), \qquad c = (0.5,\ 1,\ 1,\ 2) \tag{7}$$

Table 12. Views, the equilibrium value of each view, and view uncertainty

View	Q (ann.)	Prior PΠ (ann.)	ck	Ωkk (daily)	View st. dev. (ann.)
V1: LLY - MRK	8.0%	7.2%	0.5	4.89e-07	17.6%
V2: XOM	0.0%	6.4%	1.0	3.76e-07	15.5%
V3: GOOG - META	5.0%	-5.3%	1.0	8.94e-07	23.8%
V4: Banks - REST	3.0%	3.2%	2.0	4.84e-07	17.5%

(d) Posterior mu_{BL} and Sigma_{BL}

Using the canonical Black–Litterman formulas (Walters):

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q] \tag{8}$$

$$\Sigma_{BL} = \Sigma + [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1} \tag{9}$$

Table 13. Prior and posterior expected excess returns and volatilities (annualised)

Asset	Prior Π	μBL	Change	σ prior	σ posterior
TSLA	35.54%	34.79%	-0.74%	62.91%	62.97%
WMT	8.23%	7.72%	-0.51%	21.15%	21.17%
BAC	15.14%	13.73%	-1.41%	25.60%	25.62%
GS	20.10%	18.63%	-1.47%	28.31%	28.33%
LLY	11.90%	11.46%	-0.44%	35.23%	35.25%
MRK	4.74%	3.99%	-0.75%	24.20%	24.22%

GOOG	17.17%	18.68%	1.50%	29.57%	29.60%
META	22.51%	18.84%	-3.67%	36.18%	36.20%
AAPL	18.34%	17.50%	-0.84%	27.71%	27.74%
XOM	6.36%	3.15%	-3.21%	22.22%	22.23%
REST	14.44%	13.57%	-0.87%	15.10%	15.12%

Table 14. Posterior covariance matrix Sigma_{BL} (annualised, times 100$)

	TSLA	**WMT**	**BAC**	**GS**	**LLY**	**MRK**	**GOOG**	**META**	**AAPL**	**XOM**	**REST**
TSLA	39.65	3.45	5.67	7.71	3.04	0.64	7.15	7.69	7.70	1.29	4.78
WMT	3.45	4.48	1.45	1.67	1.43	0.53	1.07	1.97	1.78	0.34	1.22
BAC	5.67	1.45	6.57	5.61	1.38	0.75	1.92	2.69	2.42	2.11	2.39
GS	7.71	1.67	5.61	8.02	2.03	0.78	3.01	3.97	3.19	2.15	3.17
LLY	3.04	1.43	1.38	2.03	12.43	2.72	1.34	3.52	1.62	0.44	1.79
MRK	0.64	0.53	0.75	0.78	2.72	5.87	0.11	0.34	1.03	0.62	0.75
GOOG	7.15	1.07	1.92	3.01	1.34	0.11	8.76	5.06	3.74	0.35	2.47
META	7.69	1.97	2.69	3.97	3.52	0.34	5.06	13.11	4.08	0.62	3.25
AAPL	7.70	1.78	2.42	3.19	1.62	1.03	3.74	4.08	7.70	1.19	2.50
XOM	1.29	0.34	2.11	2.15	0.44	0.62	0.35	0.62	1.19	4.94	1.01
REST	4.78	1.22	2.39	3.17	1.79	0.75	2.47	3.25	2.50	1.01	2.29

The posterior moves most where our views disagree with equilibrium. The market-implied GOOG – META spread is –5.3%, and our +5% view pulls it to about –0.2%: Alphabet's expected excess return rises by 1.5 points and Meta's falls by 3.7 points. The XOM view cuts Exxon's premium from 6.4% to 3.2%. Views 1 and 4 hardly move anything, because the market was already pricing roughly what we believed (7.2% for LLY – MRK and 3.2% for banks vs the rest). This is a useful property of the model: a view only changes the answer if it is actually different from what the market already implies. Because tau is small, Sigma_{BL} is only slightly larger than Sigma (volatilities rise by 0.02–0.06 pp).

Optimal Black–Litterman weights

We computed the optimal weights in two ways. (i) The unconstrained BL portfolio over all 11 assets, $w^* = (delta Sigma_{BL})^{-1} mu_{BL}, normalised to sum to one. This shows the pure tilts the views create relative to the market. (ii) The maximum-Sharpe portfolio on the ten

investable stocks using mu_{BL} and $Sigma_{BL}$, under the same constraints as Part 1 (long-only, and long-only with the 15% cap). For reference, we also show the "prior-only" portfolio (Pi, no views), which separates the effect of the equilibrium prior from the effect of our views.

Table 15. Unconstrained Black–Litterman weights over the 11-asset market

Asset	Market weight	BL unconstrained	Tilt vs market
TSLA	2.19%	2.44%	0.25%
WMT	0.80%	0.89%	0.09%
BAC	0.63%	1.25%	0.63%
GS	0.43%	1.03%	0.60%
LLY	1.02%	2.65%	1.63%
MRK	0.35%	-1.13%	-1.48%
GOOG	2.01%	10.25%	8.24%
META	2.84%	-4.84%	-7.68%
AAPL	6.65%	7.42%	0.77%
XOM	0.86%	-10.64%	-11.49%
REST	82.24%	90.67%	8.43%

Table 16. Constrained weights. Left: long-only. Right: long-only with 15% cap

Stock	Part 1 Step 1	Prior only	BL	Part 1 Step 2	Prior only	BL
TSLA	0.0%	4.5%	5.1%	1.6%	4.9%	6.5%
WMT	43.7%	7.2%	8.1%	15.0%	8.5%	12.1%
BAC	0.0%	5.8%	5.1%	15.0%	10.6%	13.6%
GS	34.7%	22.2%	23.4%	15.0%	15.0%	15.0%
LLY	0.0%	5.7%	8.2%	14.4%	5.8%	8.3%
MRK	0.0%	5.2%	3.1%	0.0%	6.0%	6.2%
GOOG	4.9%	12.2%	23.9%	15.0%	13.5%	15.0%
META	16.7%	12.9%	3.8%	15.0%	13.3%	8.3%
AAPL	0.0%	18.6%	19.2%	4.1%	15.0%	15.0%

XOM	0.0%	5.7%	0.0%	4.9%	7.4%	0.0%

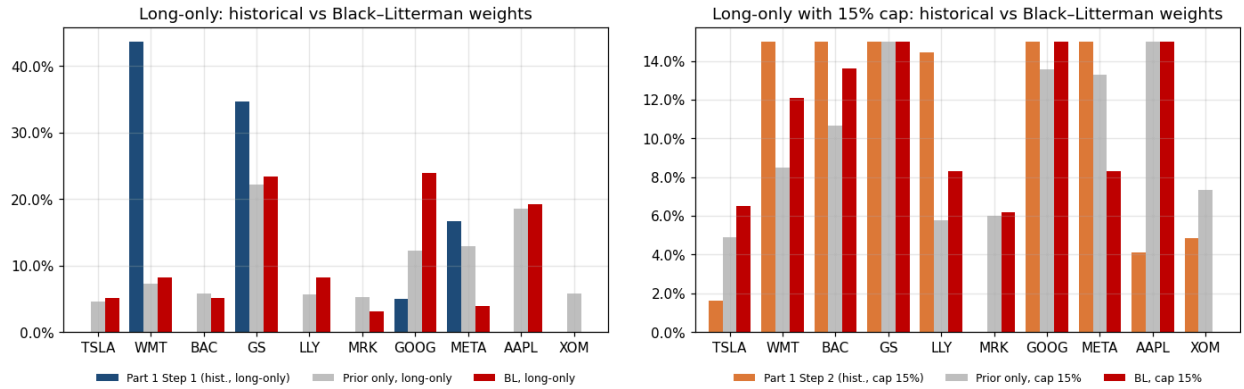


Figure 5. Historical (Part 1), prior-only and Black–Litterman weights

(e) Black–Litterman vs Part 1

The differences are large. The Part 1 long-only portfolio was 78% in two stocks (WMT and GS), while the BL long-only portfolio holds nine names, with the largest positions in Alphabet (23.9%), Goldman (23.4%) and Apple (19.2%). Most of this change comes from replacing noisy historical means with equilibrium returns, not from our views: the total absolute weight change between Part 1 and BL is 1.19 (long-only), but between the prior-only portfolio and BL it is only 0.35. The views still matter where they disagreed with the market. **View 3 had the biggest effect**, doubling GOOG (12.2% ->23.9%) and cutting META from 12.9% to 3.8%. **View 2** removed Exxon completely (5.7% -> 0%). View 1 moved LLY up (5.7% ->8.2%) and MRK down (5.2% -> 3.1%), while View 4 had almost no impact because it agreed with the prior. In the unconstrained 11-asset solution (Table 15), the views show up as clean long/short tilts around the market weights: +8.2 pp GOOG against –7.7 pp META, –11.5 pp XOM, and +1.6 pp LLY against –1.5 pp MRK.

Do weights change more or less with constraints? Less. With the 15% cap, the total weight change caused by the views falls from 0.35 to 0.25, and the change versus Part 1 falls from 1.19 to 0.44. The cap is already binding for GS, GOOG and AAPL in the BL portfolio, so the extra conviction in Alphabet cannot be expressed beyond 15%, and the optimiser has to spread the remaining weight elsewhere. Constraints therefore make the final portfolio more robust but also hide part of the manager's views. With no constraints at all, the views are expressed in full, including short positions in META, MRK and XOM. Under the BL inputs, both BL portfolios have a higher expected Sharpe ratio (0.85 and 0.84) than the Part 1 portfolios evaluated on the same inputs (0.72 and 0.80).

Table 17. Out-of-sample check including the BL portfolios, Oct–Dec 2025

Portfolio	Cumulative return	Sharpe ratio (ann.)	Max drawdown
Part 1 Step 1: long-only	6.87%	1.51	-5.02%
Part 1 Step 2: cap 15%	13.09%	3.42	-2.97%
BL long-only	15.43%	3.38	-4.09%
BL cap 15%	12.98%	3.16	-3.69%
S&P 500	2.35%	0.39	-5.11%

As an extra check, we ran the BL portfolios through the same out-of-sample quarter. The BL long-only portfolio returned 15.4% (Sharpe 3.38), the best of all, helped by being overweight Alphabet and underweight Meta, exactly as View 3 intended. The capped BL portfolio returned 13.0%, almost the same as the Part 1 capped portfolio. The Exxon view was wrong in this quarter (XOM rose 6.7%), but the position was small, so it did little damage.

2.5 Summary of key results and recommended course of action

- **The unconstrained mean-variance portfolio is concentrated** (four stocks, 78% in WMT and GS) and has the best in-sample Sharpe ratio (1.86), but it under-performed out of sample (6.9% vs 13.1% for the capped portfolio).
- **The 15% cap shortens the efficient frontier and lowers the in-sample Sharpe ratio to 1.57**, but it diversifies estimation risk and was the better portfolio in the test quarter (Sharpe 3.42, drawdown –3.0%).
- **The capped portfolio is a market-like (beta approx 1.05) large-cap portfolio** with a value tilt and an aggressive-investment tilt. Alpha is not significant, so its excess return is mostly explained by factors.
- **Simulation reaches the optimal Sharpe ratio quickly but the optimal weights only very slowly** (about 10^6 portfolios for ~2 pp accuracy, and 18 times more raw draws with a 30% cap). Optimisation should be used.
- **Black–Litterman produces diversified, intuitive weights and lets the manager express views in a controlled way.** Constraints dampen the effect of views.

Recommendation. We recommend that the client hold the **Black–Litterman portfolio with the 15% cap**, or, if the manager prefers not to rely on views, the Part 1 capped portfolio, which is very close. It respects the client's limit, holds nine of the ten stocks, has better-grounded expected returns than the purely historical portfolio, and expresses the GOOG-over-META and underweight-energy views in moderate size. The concentrated unconstrained portfolio should not be used. The portfolio should be reviewed quarterly: re-estimate Sigma, update the market weights and revisit each view, and in particular the Exxon view, which did not work in Q4 2025. Factor exposures should be monitored as well, especially the negative size and investment loadings, which would hurt if small caps or low-investment "value" stocks start to lead.

3. Non-technical report

What we found

We were asked to build the best mix of ten large US companies for the client, using their performance between September 2023 and September 2025, and then to check how it held up in the last three months of 2025.

If we simply chase the best past results, the recommended mix puts almost 80% of the money into just two companies, Walmart and Goldman Sachs. On paper this looked like the best trade-off between return and risk. But it is fragile: it depends heavily on two years of past returns repeating, and they rarely do. When the client asked that no company make up more than 15% of the portfolio, the money was spread across nine companies. On past data this looked slightly less attractive. In the following three months, however, the spread-out portfolio gained about **13%**, almost double the concentrated one (about 7%), and its worst dip was only 3% compared with 5%. Over the same period the S&P 500 rose only about 2%. The concentrated portfolio missed the big gains in Eli Lilly, Merck and Alphabet and was hurt by a fall in Meta.

We also checked what really drives the client's portfolio. Most of its ups and downs follow the overall stock market, slightly more than one-for-one. Beyond that, it behaves like a portfolio of very large companies, with a mild preference for cheaper "value" companies such as the banks, and a tilt towards companies that are investing heavily in growth, such as the tech firms building AI infrastructure. After accounting for these drivers, there is no reliable evidence of extra return beyond what the market and these styles explain.

Finally, we combined the market's own pricing with a few of our own opinions: Eli Lilly should do better than Merck, oil companies will struggle with weak oil prices, Alphabet should do better than Meta, and the big banks should do slightly better than the market. This produced a

balanced portfolio that leans more towards Alphabet and less towards Meta and Exxon, without making any big bets. In the test quarter this approach also performed well.

What we recommend

- **Invest in the diversified portfolio that respects the 15% limit**, adjusted for our market views. In practice this is roughly 15% each in Goldman Sachs, Alphabet and Apple, 12–14% in Walmart and Bank of America, about 8% in each of Eli Lilly and Meta, 6–7% in Tesla and Merck, and no Exxon.
- **Avoid the concentrated two-stock portfolio.** It looked best on past data but it is the most exposed to a single disappointment.
- **Review the portfolio every quarter**, and revise our views when the facts change. For example, if oil prices recover, the zero weight in Exxon should be reconsidered.

What will drive each portfolio

- **All portfolios**: The biggest risk is which way the US stock market goes. If the market dropped 10%, we'd expect the portfolio to drop about 10–11%.
- **Concentrated portfolio (Part 1, no limit)**: It mostly depends on just two companies. Walmart is affected by how much people spend and by tariffs on goods it imports. Goldman Sachs depends on how many deals happen, how busy trading is, and interest rates. If either company has a bad time, the whole portfolio gets hurt.
- **15%-limit portfolio**: Its risks are spread out much more. Bank profits and interest rates matter for Bank of America and Goldman Sachs, spending on AI and tech matters for Alphabet and Meta, drug approvals and prices matter for Eli Lilly, and how much people buy matters for Walmart. It would fall behind the market if smaller companies, or companies that spend less on growth, start doing better.
- **Views-based portfolio**: On top of all that, it depends on whether Alphabet's AI and cloud business does better or worse than Meta's. It's also affected by oil prices, because it holds no Exxon, so if oil prices go up it misses out on those gains. Changes to bank rules could also affect it.

4. Conclusion

Working on this project with real data showed us how much mean-variance optimisation depends on the numbers we feed into it. The "best" portfolio based on past data ended up being the most concentrated and the most likely to break. A simple limit on weights cost us a

little on the training data, but in return it did much better on new data. The factor regression showed that the client is mostly buying exposure to the overall market, with a lean towards big companies, value stocks and companies that spend a lot to grow. The OLS and robust results also agreed with each other. Random simulation can get close to the best portfolio, but only after a huge number of tries, and limits on the weights make it much worse. We would only use it to help explain things, not to make real decisions.

Out of all the methods we tried, Black–Litterman was the most useful in practice. It starts from the market, only changes the parts where our views are different, and keeps the portfolio well spread out, especially with the client's 15% limit. If we had to sum up what we learned, it would be two things. First, mistakes in our estimates are a bigger problem than which model we pick, so limits that keep the portfolio diversified are worth the small cost. Second, our views should be few, backed by good reasons, and given more or less weight depending on how confident we are in them.

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